CS315 · ALGORITHMS ANALYSIS & DESIGN

Midterm 472 Practice

From L1 to L7

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01 Questions

Q1 MATRIX CHAIN

Apply the Dynamic programming algorithm and give the optimal solution to the matrix chain multiplication ABCD where the order for matrix:

A = 4 × 5 B = 5 × 8 C = 8 × 3 D = 3 × 7

Minimum operations = _____ Best order of multiplication: _____

	A	B	C	D
A	0	160		
B		0	120	
C			0	
D				0

Q2 TRUE / FALSE + MCQ

1. The code length in Huffman codes depend on the frequency of occurrence of characters.

○ True ○ False

2. Greedy algorithm can solve all the dynamic programming problems.

○ True ○ False

3. The time to run MERGESORT on a list of size n, $T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + T(MERGE(\frac{n}{2}))$

○ True ○ False

4. In order to compare two algorithms, the same hardware and software environments must be used without any consideration of inputs type and its size.

○ True ○ False

5. Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment.

○ True ○ False

6. An algorithm which does NOT use the past results and uses them to find new results is:

a. Greedy Method b. Divide and conquer c. Dynamic programming d. None of the mentioned

Q3 LCS

Using the dynamic programming approach, find the longest common subsequence between two strings:

X = ACBAED and Y = ABCADF.

Length _____ The subsequence: 1 _____ 2 _____

	⌀	A	B	C	A	D	F
⌀							
A							
C							
B							
A							
E							
D							

Q4 HASH TABLE

Insert integer keys A = {4, 13, 49, 6, 40, 33} in order into a hash table of size 9 using the hash function h1(k) = k mod 9. Collisions should be resolved via chaining, where collisions are stored at the end of a chain. Fill in the hash table after all keys have been inserted.

0	1	2	3	4	5	6	7	8

Q5 B-TREE

A) Create a B-tree with order 5, min degree 2 the following item [24, 52, 90, 40, 11, 86, 61, 50]

B) insert 29

C) delete 52, 11